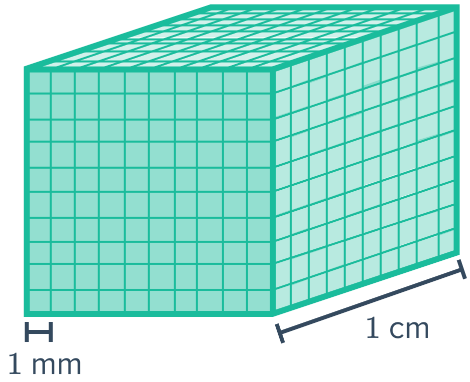
Units of volume and capacity

1 The cube shown in the diagram has a volume of $1\ \text{cm}^3$. Find its volume in cubic millimetres.



2 Convert the following to cubic millimetres:

a $3.5\ \text{cm}^3$	b $4.2\ \text{m}^3$	c $4\ \text{m}^3$	d $19\ \text{cm}^3$

3 Convert the following to cubic centimetres:

a $47\,600\ \text{mm}^3$	b $8.12\ \text{m}^3$	c $7000\ \text{mm}^3$	d $33\ \text{m}^3$
e $1\ \text{mL}$	f $8000\ \text{mL}$	g $40\ \text{mL}$	h $4.47\ \text{mL}$

4 What operation is required to convert cubic centimetres to cubic metres?

5 Convert the following to cubic metres:

a $1\,070\,000\,000\ \text{mm}^3$	b $51\,980\,000\ \text{cm}^3$	c $460\,000\,000\ \text{mm}^3$	d $16\,000\,000\ \text{cm}^3$
e $6000\ \text{L}$	f $800\ \text{L}$	g $18\ \text{kL}$	h $0.95\ \text{kL}$

6 Which metric unit represents 'one thousandth of a litre'?

7 Which metric unit represents 'one thousand millilitres'?

8 Convert the following to millilitres:

a $6\ \text{L}$	b $1.54\ \text{L}$	c $2\ \text{kL}$	d $0.17\ \text{kL}$
e $1\ \text{cm}^3$	f $8.5\ \text{cm}^3$	g $9000\ \text{mm}^3$	h $0.005\ \text{m}^3$

9 Convert the following to litres:

a 4920 mL b 5000 mL c 9 kL d 1.67 kL

e 18.62 m^3 f 2 m^3 g 6750 cm^3 h 800 cm^3

10 Convert the following to kilolitres:

a 6570 L b 84.6 L c $1\,500\,000 \text{ mL}$ d $60\,000 \text{ mL}$

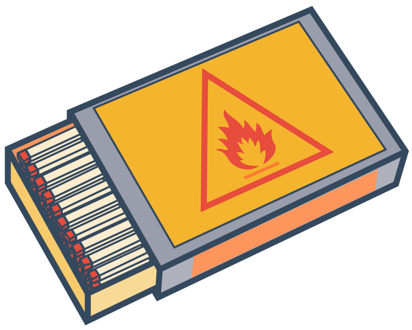
e 17 m^3 f 0.37 m^3 g $25\,000\,000 \text{ cm}^3$ h $54\,000 \text{ cm}^3$

Volume estimation

11 Consider the matchbox shown:

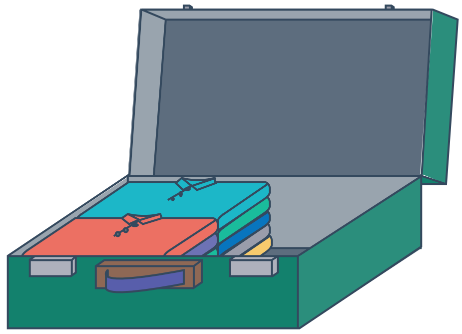
a State the most appropriate unit for measuring the volume of a match box.

b Estimate the volume of the matchbox.



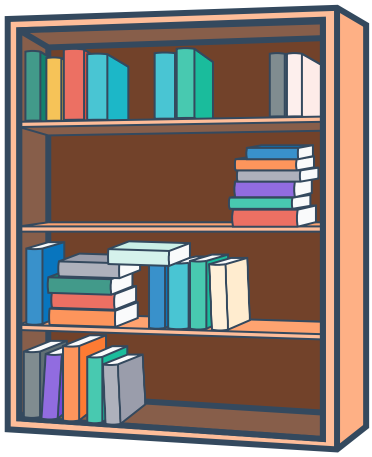
12 The volume of the suitcase shown has a total capacity of $88\,500 \text{ cm}^3$.

Estimate the volume of the contained shirts.



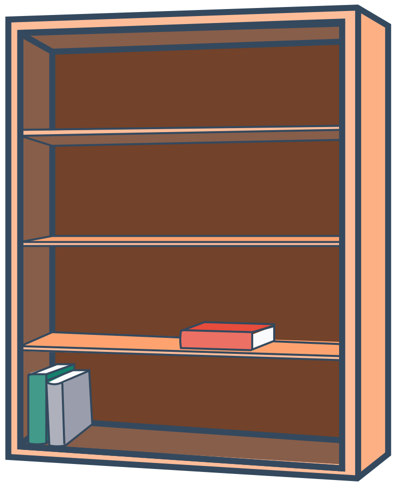
13 The volume of the bookshelf shown below is known to be $0.64\ \text{m}^3$.

Estimate the volume of the books on the bookshelf.



14 The volume of the bookshelf shown is known to be $0.68\ \text{m}^3$.

Estimate the volume of the books on the bookshelf.

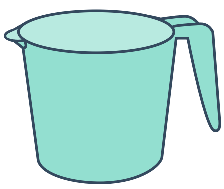


15 State the most appropriate unit for measuring the volume of the following:

a Sand pit
b Wardrobe
c Water glass
d Swimming pool
e Air in a hall
f Water bottle
g Air in a balloon

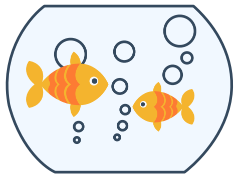
Capacity estimation

16 Which units can we measure with a jug like the one in the picture?

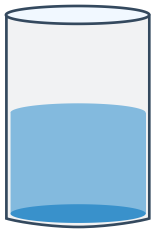


17 State the most appropriate unit for measuring the capacity of a cup of coffee.

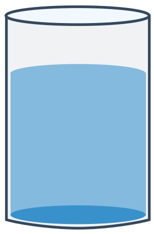
18 State the most appropriate unit for measuring the capacity of a fish bowl.



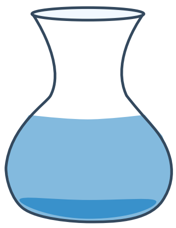
19 Estimate the volume of liquid in this container if it is able to hold $2400\ \mathrm{mL}$ when full.



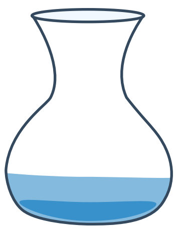
20 Estimate the volume of liquid in this container if it is able to hold $3.6\ \mathrm{L}$ when full.



21 Estimate the volume of liquid in this container if it is able to hold $3600\ \mathrm{mL}$ when full.

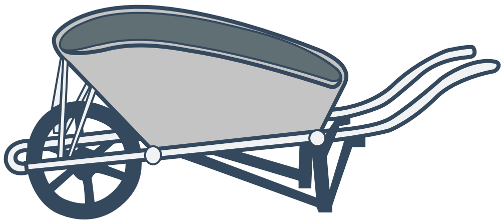


22 Estimate the volume of liquid in this container if it is able to hold 3.8 litres when full.



23 A landscaper needs to move soil across the garden. Choose the best estimate for the capacity of soil that can be carried by a wheelbarrow in one trip.

A 771 mL **B** 21 ML **C** 71 L



24 State the most appropriate unit of measure for finding the volume of the following:

a Water in a dam.

b A dose of vaccine injection.

c A small carton of milk.

d The Atlantic Ocean.

e A dosage of medicine.

f A bottle of washing detergent.

Applications

25 Quiana bought a 400 mL bottle of soft drink. How many litres of soft drink did she buy?

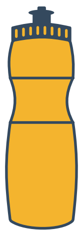
26 The following baby bottle holds 100 mL of milk:

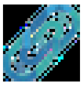
Find its volume in cm^3.

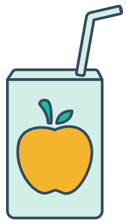
27 The following water bottle has a volume of 700 cm^3:

a Find its capacity in millilitres.

b Find its capacity in litres.



28 This juice box contains $1600 \mathrm{~mL}$ of juice: 



a Find its volume in cm^3.

b Calculate the length of the juice box if it is $8 \mathrm{~cm}$ tall and $5 \mathrm{~cm}$ wide.

29 A cubic box has capacity of $343 \mathrm{~L}$. Find its volume in cubic centimetres.

30 A small pond contains $3900 \mathrm{~L}$ of water. Find the capacity of the pond in cubic metres.

31 A jug has a volume of $12\,395 \mathrm{~cm}^3$. How many litres of soda water can it hold? Round your answer to three decimal places.